Fiscal Year:	FY 2019	**Task Last Updated:**	FY 01/18/2019
PI Name:	Marshall-Goebel, Karina Ph.D.		
Project Title:	Assessment of Artificial Gravity As a Countermeasure to Cephalad Fluid Shifting		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Karina.j.marshallgoebel@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	281-792-9996
Organization Name:	NASA Johnson Space Center		
PI Address 1:	Mail Code SK111		
PI Address 2:	2101 NASA Parkway		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:	New affiliation as of spring 2022: NASA Johnson Space Center; Human Physiology, Performance, Protection & Operations (H-3PO) Laboratory New affiliation as of fall 2018: KBR/NASA Johnson Space Center, Cardiovascular and Vision Laboratory, Houston; previously at Massachusetts General Hospital		
Project Type:	Ground	**Solicitation / Funding Source:**	2017 TRI-RFA-17-01: Translational Research Institute for Space Health (TRISH) Postdoctoral Fellowships
Start Date:	03/01/2018	**End Date:**	12/31/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2018 per TRISH (Ed., 5/29/2020) NOTE: End date changed to 10/31/2018 per E. Urquieta/TRISH (Ed., 12/17/18) NOTE: End date changed to 9/30/2018 (originally 2/28/2020) per PI move to NASA Johnson Space Center, per TRISH (Ed., 10/25/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Zhang, Quan Ph.D. (MENTOR: Massachusetts General Hospital)		
Grant/Contract No.:	NNX16AO69A-P0203		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	POSTDOCTORAL FELLOWSHIP Despite ongoing research efforts to elucidate the etiology of the Spaceflight Associated Neuro-ocular Syndrome (SANS) and develop potential countermeasures, visual impairment remains a critical risk factor for long-duration human spaceflight. A growing body of evidence suggests that microgravity-induced cephalad fluids and redistribution of blood volume, especially within the intracranial space, may play a major role in the development of SANS. The goal of this proposal is to assess the physiological changes related to fluid shifts during a ground-based spaceflight analog and to assess artificial gravity (AG) via short-arm human centrifugation as a countermeasure against cephalad fluid shifting and ultimately for SANS. Thus, we propose to 1) determine the physiological effects of gravitational vectors through various body positions including head-up tilt, supine and head-down tilt on body fluid distribution as well as cerebral hemodynamic and cardiovascular physiology using novel non-invasive technologies, 2) determine the effects of short-arm centrifugation on cerebral hemodynamics and full body fluid distribution, and 3) utilize multiple experimental platforms to quantify the ability of AG to reinstate supine and upright fluid distributions as a countermeasure to cephalad fluid shifting.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Currently, there is no countermeasure against the physiological and anatomical neuro-ocular effects of cephalad fluid shifting during spaceflight. Microgravity-driven fluid redistribution within the intracranial space and an increase in intracranial pressure (ICP) relative to the upright position on Earth are assumed to be major factors in the development of SANS. A countermeasure that can successfully redistribute body fluids similar to the upright position on Earth is thus expected to be important for the prevention or mitigation of SANS. In addition, the ability to image and monitor whole body fluid distribution is critical to safe and successful exploration class missions and human spaceflight. The current study tested the ability of short-arm human centrifugation to reinstate a head-to-foot physiological gradient similar to the upright position on Earth and redistribute blood volume to the lower limbs. Centrifugation up to 2-g at the feet (1-g at the heart) immediately re-distributed blood volume from the head and chest to the legs within minutes and sustained the volume shift during the entire duration of centrifugation. Centrifugation has previously been found to be an effective countermeasure against other physiological system deconditioning. As centrifugation may be able to counteract the headward fluid shift in space hypothesized to underlie SANS in space, artificial gravity may represent the first multi-physiological system countermeasure for spaceflight. In addition, this research study utilized and tested novel non-invasive technologies that have been developed for use on Earth that also can provide benefits to monitor and maintain health in space. The research study also helped determine the ability of these novel technologies to function in an extreme environment (i.e., hypergravity) in an operator independent fashion. The fellowship award allowed the postdoctoral fellow to identify, investigate, and test several physiological monitoring techniques to characterize the physiological response to gravitational loading, with an outlook of finding a countermeasure for SANS during spaceflight. The physiological monitoring devices were tested during exposure to various gravitational loads to determine proper function of all devices during centrifugation, technological limitations, compatibility, and optimization of signals during subject movement and altering g-loads. In addition, feasibility assessments were made on the ability to use and operate all devices remotely from the centrifuge control room.
Task Progress:	As humans are adapted to live in a 1-g environment, exposure to reduced gravity levels, including microgravity, affects the homeostatic physiology of many body systems. Recently, it has been determined that ocular changes affect approximately half of astronauts during long-duration spaceflight missions on-board the International Space Station (ISS), resulting in decreased visual acuity. This Spaceflight Associated Neuro-ocular Syndrome (SANS) is associated with a myriad of functional and structural neuro-ophthalmic changes including optic disc edema, chorioretinal folds, retinal nerve fiber layer thickening, decreased near visual acuity, and posterior eye globe flattening. With the removal of all gravitational vectors in space, and thus physiological hydrostatic pressure gradients, a headward fluid shift occurs. Microgravity-driven fluid redistribution within the intracranial space and an increase in ICP relative to the upright position on Earth may be major factors in the development of SANS. A countermeasure that can successfully redistribute body fluids similar to the upright position on Earth is critical to the prevention of SANS. The objective of this Postdoctoral Fellowship was to assess the physiological changes related to fluid shifts and to assess AG as a potential countermeasure against cephalad fluid shifting, with the following specific aims: Aim 1: Determine the physiological effects of gravitational vectors through various body positions including head-up tilt, supine and head-down tilt on body fluid distribution as well as cerebral hemodynamic and cardiovascular physiology using novel non-invasive technologies. Aim 2: Determine the effects of short-arm centrifugation on cerebral hemodynamics and full body fluid distribution. Aim 3: Utilize multiple experimental platforms to quantify the ability of AG to reinstate supine and upright fluid distributions as a countermeasure to cephalad fluid shifting. As the overall goal of postdoctoral fellowships is to foster the development of an independent investigator, in this case in aerospace medicine and physiology, this fellowship provided training in 1) spaceflight research specific methodologies, 2) the use of non-invasive cerebral and cardiovascular technologies compatible with the spaceflight environment, 3) analysis techniques for continuously recorded physiological data, and 4) training on countermeasure techniques relevant to the space medicine domain. Several of the proposed non-invasive devices have not yet been tested in an extreme environment setting, such as during hypergravity exposure. The eventual use of non-invasive technologies for spaceflight missions will require them to be both easily operated and able to withstand varying environmental stressors. This research project allowed for a feasibility analysis of the implemented non-invasive technologies during hypergravity including technological limitations, compatibility, and optimization of signals, which may give insights into their suitability for use in spaceflight in the current design. Results from human testing on the short-arm human centrifuge demonstrated that centrifugation up to 2g at the feet produced significant fluid shifting to the lower limbs. Notably, near-infrared spectroscopy (NIRS) was utilized to continuously measure cerebral perfusion, blood oxygenation, and hemodynamics as well as blood volume in multiple body segments (head, chest, upper thigh, and calf). Significant shifting of blood volume from the upper body (head and chest) to the lower body (thigh and calf) was seen immediately upon ramp-up of the short-arm human centrifuge. The shifting of blood reached a peak after about 1.5 min of 2g centrifugation (at the feet) and this shift was sustained for the duration of the 2g centrifugation. Immediately upon ramp-down of the centrifuge, blood volume began to shift back towards the chest and head and upon cessation of the centrifuge, the blood volume returned to pre-centrifugation baseline. In addition, internal jugular vein (IJV) dimensions were monitored continuously during centrifugation. The IJVs are the primary outflow pathway of the cerebral venous system in the supine posture on Earth,

	and venous congestion in space has been hypothesized to be a major factor in SANS. Thus, assessment of the IJV provides a direct measure of cerebrovascular outflow status. Similar to the changes in blood volume seen with NIRS, the IJV displayed immediate and significant changes in cross-sectional area during centrifugation. From supine rest, the area of the IJV decreased 56% during 0.5-g exposure, 96% during 1-g exposure, and 94% during 2-g exposure. This change in IJV area was sustained during the entire duration of centrifugation exposure. Similar to the blood volume distribution measured with NIRS, IJV area immediately returned to pre-centrifugation baseline upon cessation of the short-arm human centrifuge. Overall, artificial gravity via short-arm human centrifugation proved to be an effective countermeasure in shifting blood volume from the upper body to the lower body. In addition, direct effects on cerebral venous outflow were seen immediately upon exposure to centrifugation, likely due to a reduction in venous return and shifting of venous blood to the lower limbs. Interestingly, significant shifting of blood was seen with just 1-g at the feet, with minimal additional shifting beyond 1-g. However, the vascular effects of centrifugation only lasted as long as the countermeasure was applied. The Postdoctoral fellowship was ended early, as the candidate received and accepted a job offer as a Senior Scientist in the NASA Johnson Space Center (JSC) Cardiovascular and Vision Laboratory to conduct SANS-related research. The planned 60-day bed rest study with and without exposure to centrifugation as a countermeasure will still be conducted in 2019; however, fellow has arranged to remain involved and plans to continue pursuing the goals of the study in her new position.
Bibliography Type:	Description: (Last Updated: 11/19/2025)
Articles in Peer-reviewed Journals	Marshall-Goebel K, Damani R, Bershad EM. "Brain physiological response and adaptation during spaceflight." Neurosurgery. 2019 Nov 1;85(5):E815-E821. https://doi.org/10.1093/neuros/nyz203 ; PMID: 31215633 , Nov-2019
Books/Book Chapters	Marshall-Goebel K. "Neuro-ocular effects of spaceflight." in "Handbook of Space Pharmaceuticals. " Ed. Y.V. Pathak, M. Araújo dos Santos, L. Zea. Cham, Switzerland: Springer, 2022. p. 261-71. https://doi.org/10.1007/978-3-030-05526-4_1 , Apr-2022